

WI4014TU – Numerical Analysis for PDE's

Lecture 1

Course Structure and Mathematical Preliminaries

Course Structure

- 1 16 weeks, 1 lecture per week
- 2 5 assignments, approximately 4 weeks per assignment
- 3 **Deadlines for each assignment, see Brightspace. Late uploads will result in point deductions**
- 4 **Identical parts of reports and codes will be considered plagiarism, and points will be deducted**
- 5 Assignments should be typeset in LaTeX (use Overleaf), converted to PDF and uploaded on Brightspace
- 6 Python code should also be uploaded
- 7 Adhere to the code structure suggested in the assignment
- 8 Feedback and grade per assignment
- 9 Deductions are made for all mathematical errors, no matter how small, and for the missing derivations and discussions
- 10 Course grade is the average grade of all 5 assignments, minimum 5.8

Course materials

- ① Main book: J. van Kan, A. Segal, and F. Vermolen, *Numerical Methods in Scientific Computing*, TU Delft Open Textbook, 2023, <https://textbooks.open.tudelft.nl/textbooks/catalog/book/70>
- ② Basic numerical analysis book: C. Vuik, F.J. Vermolen, M.J. Vuik, *Numerical Methods for ordinary differential equations*, TU Delft Open Textbook, 2023
<https://textbooks.open.tudelft.nl/textbooks/catalog/book/57>
- ③ Calculus: J. Stewart, *Calculus: Early Transcendentals*
- ④ Assignments 1–5 (Brightspace)
- ⑤ Additional reading: *The FEniCS Tutorial*, <https://fenicsproject.org/tutorial/>
- ⑥ These slides are linked to at Brightspace
- ⑦ Install a working Python 3 environment with `numpy`, `scipy`, `matplotlib`, FEniCS package <https://fenicsproject.org>
- ⑧ Consider using Linux (virtual environment or dual-boot)
- ⑨ LaTeX environment (Overleaf)

Creating a numerical PDE solver

- 1 Get your physics, PDE's, and BC's right. Decide 1D, 2D, or 3D?
- 2 Choose between FD, FV, and FE methods
- 3 Discretization on paper (required details depend on software choices):
matrix elements (FDM, FVM); weak form of PDE's, choice of basis/test functions (FEM)
- 4 Software implementation:
 - ▶ Choose: package (COMSOL, ANSYS), high-level library (FEniCS, Matlab), low-level library (OpenFOAM), from scratch (Python, Matlab, C/C++, FORTRAN)
 - ▶ Geometry, materials, mesh, refinement (use software tools)
 - ▶ System matrices (automated in packages, difficult from scratch)
 - ▶ Non-linear and/or time-domain solver, linear-algebraic solver, preconditioning (use existing libraries)
- 5 Debug, calibrate, compare with others (difficult, tedious)
- 6 Study something new, make predictions (have fun)

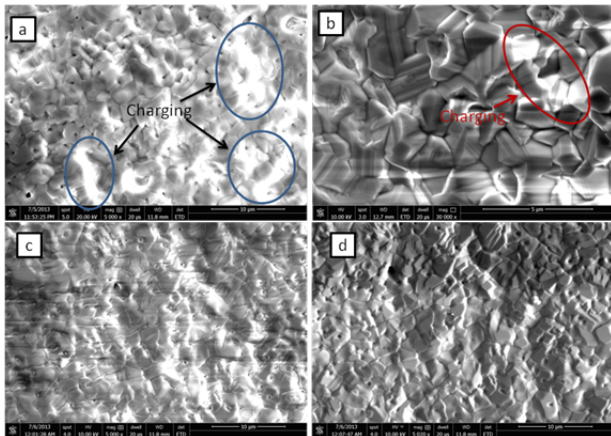
Numerical Analysis of PDE's: A Case Study

Charge injection and transport in insulating materials

B Raftari, NV Budko, C Vuik, Self-consistent drift-diffusion-reaction model for the electron beam interaction with dielectric samples *Journal of Applied Physics*, 118 (20), 2015.

B Raftari, N Budko, K Vuik, A modified and calibrated drift-diffusion-reaction model for time-domain analysis of charging phenomena in electron-beam irradiated insulators *AIP Advances*, 8 (1), 2018.

Understand the problem



Write down your PDE's

$$\partial_t n + \nabla \cdot \mathbf{J}_n = G_n - R_n + S_n$$

$$\partial_t p + \nabla \cdot \mathbf{J}_p = G_p - R_p + S_p$$

$$\mathbf{J}_n = -D_n \nabla n + \mu_n n \nabla V$$

$$\mathbf{J}_p = -D_p \nabla p - \mu_p p \nabla V$$

$$\nabla \cdot (\varepsilon \nabla V) = -\frac{e}{\varepsilon_0} (p + C - n - n_T)$$

$$\partial_t n_T = (R_n - G_n) - (R_p - G_p)$$

$$R_n = \sigma_n v_{th} n (N_T - n_T), \quad R_p = \sigma_p v_{th} p n_T$$

$$G_p = \sigma_p v_{th} n_i (N_T - n_T), \quad G_n = \sigma_n v_{th} n_i n_T$$

$$S_{n/p} = g_{n/p}(\mathbf{x}) f(t)$$

Specify initial and boundary conditions

- At $t = 0$: $n_0 = p_0 = n_i$ and $n_T = N_T/2$ (steady-state)
- Isolated interfaces: $\mathbf{J}_n \cdot \boldsymbol{\nu} = \mathbf{J}_p \cdot \boldsymbol{\nu} = 0$ (no current)
- Ohmic contacts: $n = p = n_i$
- At least at one interface: $V = V_0$ (and fully coupled approach)
- Sample-vacuum interface? Assumed isolated in semiconductor device modeling

Sometimes, invent your own boundary conditions

Sample-vacuum interface with tertiary current:

$$\mathbf{J}_n \cdot \boldsymbol{\nu} = \begin{cases} v_e(n - n_i) + \alpha \frac{\partial V^-}{\partial \vec{\nu}}, & n > n_i \\ 0, & \text{otherwise} \end{cases}$$
$$\mathbf{J}_p \cdot \boldsymbol{\nu} = 0$$

Spatial distribution is governed by the attractive force:

$$\left. \frac{\partial V^-}{\partial \vec{\nu}} \right|_{\Sigma} = \begin{cases} \left. \frac{\partial V}{\partial \vec{\nu}} \right|_{\Sigma}, & \text{if } \frac{\partial V}{\partial \vec{\nu}} < 0 \\ 0, & \text{otherwise} \end{cases}$$

Choose discretization method, solver, software

- ① Finite Volume or Finite Element Methods are better for this type of problems
- ② Finite Volume: OpenFOAM
 - ▶ Advantages: free, powerful, fast
 - ▶ Disadvantages: poorly documented, did not support specific boundary conditions at the time, need to be good at C++
- ③ FEM: COMSOL
 - ▶ Advantages: well-documented, supports general BC's, adaptive grids, GUI-based, Matlab-type language
 - ▶ Disadvantages: commercial, not flexible, memory hungry

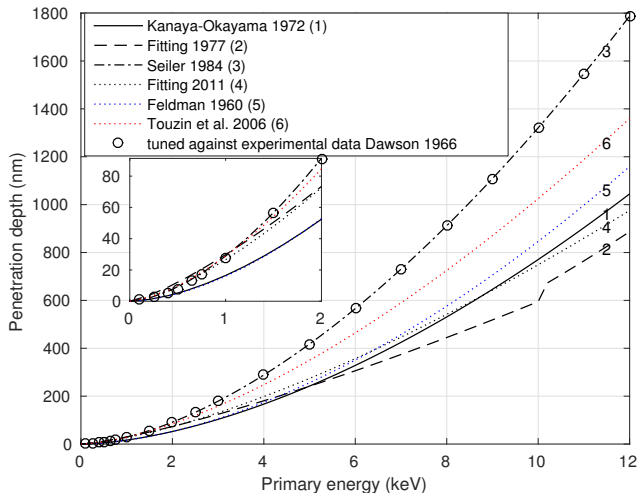
Software implementation

- We skip it for the moment...

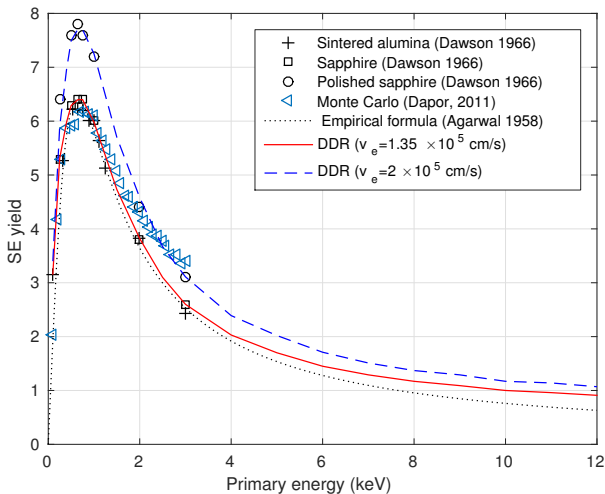
Calibration: search literature for material parameters

Parameter	SiO ₂	Al ₂ O ₃	Unit
ε	3.9	10	
μ_n	20	4	cm ² V ⁻¹ s ⁻¹
μ_p	0.01	0.002	cm ² V ⁻¹ s ⁻¹
σ_n	10 ⁻¹⁵	10 ⁻¹⁵	cm ²
σ_p	10 ⁻¹⁷	10 ⁻¹⁷	cm ²
v_{th}	10 ⁷	10 ⁷	cm s ⁻¹
ρ	2.2	3.98	g cm ⁻³
E_i	28	28	eV
N_t	3×10^{19}	3×10^{19}	cm ⁻³
v_e	0.8×10^5	$\begin{cases} 1.35 \times 10^5 & \text{unpolished} \\ 2 \times 10^5 & \text{polished} \end{cases}$	cm/s

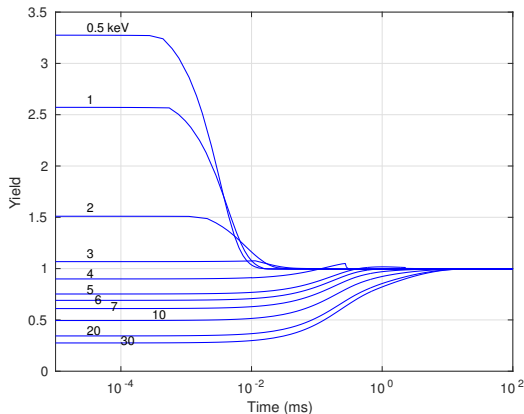
Calibration: tune unknown/uncertain parameters against experimental data



Calibrated results: standard yield, sapphire

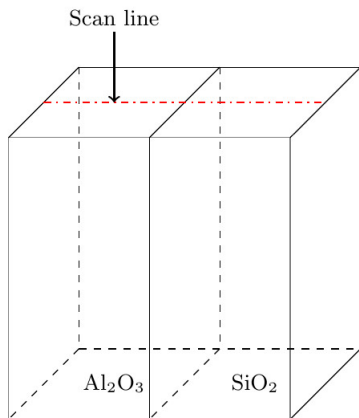


Compare with others: yield evolution for amorphous alumina

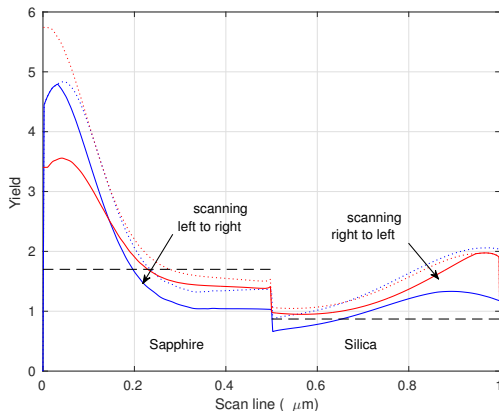


Agrees with 1D Flight-Drift model of Touzin et al 2006 (beam current 10^{-5} A/cm^2)

Studying a new configuration: scanning of inhomogeneous samples

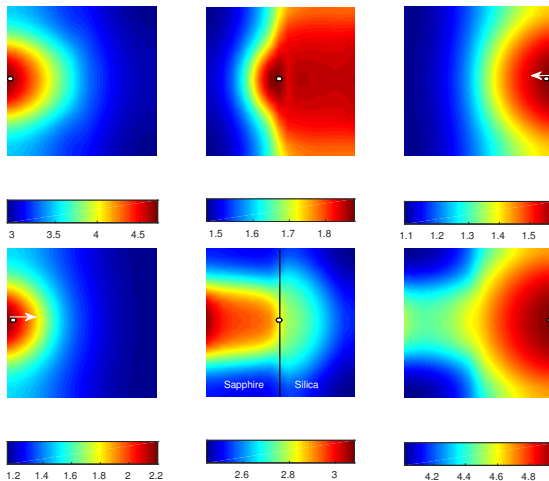


Making conclusions: image depends on the beam current, scanning speed and direction, etc.



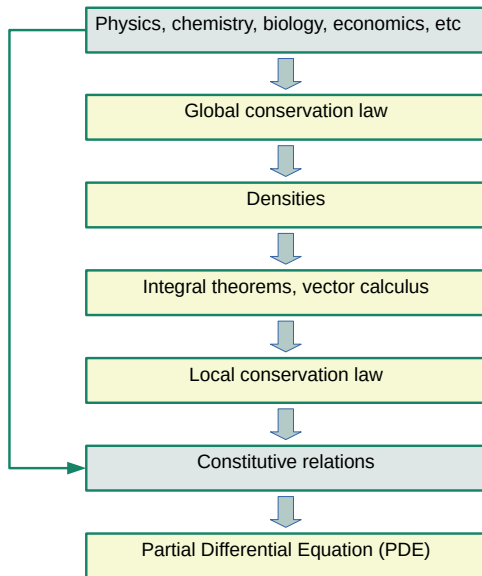
Yield evolution during dynamic scanning at 5 keV, 10 pA

Demonstrating/explaining a new/strange effect: charging depends on the direction of scanning



Mathematical Preliminaries

Derivation of PDE's



Gradient

Given a scalar function $u(x, y, z)$ of three variables, the *gradient* of u is the vector:

$$\text{grad } u = \frac{\partial u}{\partial x} \vec{i} + \frac{\partial u}{\partial y} \vec{j} + \frac{\partial u}{\partial z} \vec{k} = \begin{bmatrix} \frac{\partial u}{\partial x} \\ \frac{\partial u}{\partial y} \\ \frac{\partial u}{\partial z} \end{bmatrix}$$

Alternatively, using the *nabla* operator ∇ the gradient of u is:

$$\nabla = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix}, \quad \text{grad } u = \nabla u$$

Directional Derivative and Gradient

Directional derivative $D_{\vec{v}}u$ of the function u in the direction of the unit vector $\vec{v} = v_x \vec{i} + v_y \vec{j} + v_z \vec{k}$, $|\vec{v}| = 1$ is defined as:

$$D_{\vec{v}}u = v_x \frac{\partial u}{\partial x} + v_y \frac{\partial u}{\partial y} + v_z \frac{\partial u}{\partial z} = \nabla u \cdot \vec{v} = |\nabla u| |\vec{v}| \cos \theta = |\nabla u| \cos \theta$$

Hence, the maximum value of the directional derivative $D_{\vec{v}}u$ is $|\nabla u|$ and it occurs when $\vec{v}$ has the same direction as ∇u .

Thus, the gradient vector ∇u gives both the direction of the maximum rate of growth of $u(x, y, z)$ at the point (x, y, z) and the magnitude of this rate.

Applications of gradient: heat flow

Fourier's law: Heat flows from warm to cold, the stronger is the temperature difference, the faster is the flow. Mathematically: Heat flux density $\vec{q}$ is proportional to the negative temperature gradient:

$$\vec{q} = -k \nabla T$$

$\vec{q}$ is the local heat flux density [W m^{-2}],
 k is material's thermal conductivity [$\text{W m}^{-1}\text{K}^{-1}$],
 ∇T is the temperature gradient [K m^{-1}],
 $T(x, y, z)$ is the local temperature [K]

Gradient in electrostatics, hydrostatics and diffusion

Electrostatic field: The electrostatic field $\vec{E}$ is defined as the negative of the gradient of the electrostatic potential ϕ , i.e., $\vec{E} = -\nabla\phi$. If ϕ is in Volts, what are the units of $\vec{E}$? The electric displacement field (electric flux density) $\vec{D}$ is defined as $\epsilon\vec{E}$, where ϵ , [F m^{-1}] is the dielectric permittivity of the material. What is the expression of $\vec{D}$ in terms of ϕ ?

Darcy's law: the (slow) flow of water through porous medium is driven by the hydrostatic pressure and depends linearly on the height of the ground water level h . What is the mathematical expression for the water flow density $\vec{q}$ in terms of h ?

Fick's law: a substance diffuses from higher to lower concentrations at a rate proportional to the concentration difference. Write diffusion flow density $\vec{q}$ in terms of the local substance concentration c ? What are the dimensions of all quantities in SI units?

Divergence

Given a vector field $\vec{U}(x, y, z) = U_x(x, y, z)\vec{i} + U_y(x, y, z)\vec{j} + U_z(x, y, z)\vec{k}$, the *divergence* of $\vec{U}$ is the scalar:

$$\operatorname{div} \vec{U} = \frac{\partial U_x}{\partial x} + \frac{\partial U_y}{\partial y} + \frac{\partial U_z}{\partial z} = \nabla \cdot \vec{U}$$

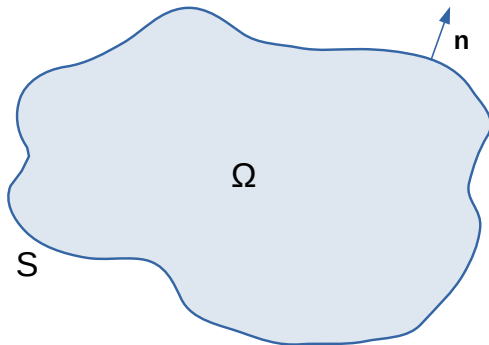
Scalar functions U_x , U_y , and U_z are called *components* of the vector field $\vec{U}$. Sometimes denoted as U_1 , U_2 , and U_3 .

In books and papers vectors and vector fields are denoted with bold font:

$$\vec{U} = \mathbf{u}, \quad \text{or} \quad \mathbf{q} = -k\nabla T$$

This notation will be used from now on.

Domain Ω , boundary surface S , normal vector $\mathbf{n}$



Flux of a vector field

- We exclude nonorientable surfaces such as the Möbius strip and consider only two-sided surfaces
- If it is possible to define a normal vector $\mathbf{n}$ at every point of S so that $\mathbf{n}$ varies continuously over S , then S is called **oriented surface** and the given choice of $\mathbf{n}$ provides S with an **orientation**

Definition. If $\mathbf{F}$ is a continuous vector field defined on an oriented surface S with unit normal vector $\mathbf{n}$, then the **surface integral of $\mathbf{F}$ over S** is

$$\Phi = \iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S (\mathbf{F} \cdot \mathbf{n}) dS$$

Φ is called the **flux of $\mathbf{F}$ across S** , and $\mathbf{F}$ is often called the **flux density**.

Applications of fluxes: Gauss's law, heat flow

Gauss's law: the net charge Q enclosed by a closed surface S is given by

$$Q = \varepsilon_0 \iint_S \mathbf{E} \cdot d\mathbf{S}$$

Heat flow: From the Fourier law $\mathbf{q} = -k\nabla T$. The rate of heat flow Q across the surface S is

$$\Phi = \iint_S \mathbf{q} \cdot d\mathbf{S} = - \iint_S k\nabla T \cdot d\mathbf{S}$$

Global form of conservation laws

Let $Q(t)$ be the total amount of some quantity inside the domain Ω at time t . There are two reasons for Q to change with time: the outward flux Φ across the boundary S of Ω and the creation or destruction of this quantity at the rate $R(t)$ inside Ω .

Then, the rate of change of Q inside Ω can be written as:

$$\frac{dQ}{dt} = -\Phi + R$$

This is the general **global** form of the **conservation law**.

Divergence Theorem

Divergence Theorem. Let Ω be a simple solid region and let S be a boundary surface of Ω with positive (outward) orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous partial derivatives on an open region that contains Ω . Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_{\Omega} (\nabla \cdot \mathbf{F}) dV$$

In words: the flux of $\mathbf{F}$ across the boundary surface of Ω is equal to the triple integral of the divergence of $\mathbf{F}$ over Ω .

Local form of conservation laws

Let $\rho(x, y, z, t)$ be the **density** of some quantity and $f(x, y, z, t)$ be the local rate of its creation/destruction. Then,

$$Q(t) = \iiint_{\Omega} \rho \, dV, \quad R(t) = \iiint_{\Omega} f \, dV$$

If $\mathbf{J}$ is the flux (current) density, then the conservation law is

$$\begin{aligned} \frac{d}{dt} \iiint_{\Omega} \rho \, dV &= - \iint_S \mathbf{J} \cdot d\mathbf{S} + \iiint_{\Omega} f \, dV \\ &= - \iiint_{\Omega} (\nabla \cdot \mathbf{J}) \, dV + \iiint_{\Omega} f \, dV \end{aligned}$$

If this holds for any Ω , then the **local conservation law** (continuity law) is

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{J} = f$$

Diffusion equation

As we know, there is a **constitutive relation** $\mathbf{J}_{\text{diffusion}} = -k\nabla\rho$, which connects $\mathbf{J}$ and ρ via a constitutive parameter of the medium k . Then, the conservation law becomes a PDE:

$$\frac{\partial\rho}{\partial t} - \nabla \cdot (k\nabla\rho) = f$$

which is called the **diffusion equation**.

Advection equation

Convection (advection) current density has the form:

$$\mathbf{J}_{\text{convection}} = \rho \mathbf{v}$$

where $\mathbf{v}$ is a given velocity field. Then, the local conservation law yields the PDE:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = f$$

which is called the **advection equation**. Since $\nabla \cdot (\rho \mathbf{v}) = \mathbf{v} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{v}$, in the case where $\nabla \cdot \mathbf{v} = 0$ (incompressible flow, solenoidal $\mathbf{v}$) we get a simpler advection equation

$$\frac{\partial \rho}{\partial t} + \mathbf{v} \cdot \nabla \rho = f$$

Convection-Diffusion equation

If current density contains both the advection (convection) and the diffusion contributions:

$$\mathbf{J} = \mathbf{J}_{\text{diffusion}} + \mathbf{J}_{\text{convection}}$$

Then, the conservation law becomes

$$\frac{\partial \rho}{\partial t} - \nabla \cdot (k \nabla \rho) + \nabla \cdot (\rho \mathbf{v}) = f$$

which is called the advection-diffusion or **convection-diffusion equation**.
In the incompressible case:

$$\frac{\partial \rho}{\partial t} - \nabla \cdot (k \nabla \rho) + \mathbf{v} \cdot \nabla \rho = f$$

Poisson equation, Laplace's operator

In the steady state

$$\frac{\partial \rho}{\partial t} = 0$$

and the diffusion equation reduces to the **Poisson equation**:

$$-\nabla \cdot (k \nabla \rho) = f$$

If also $k \neq 0$ is a constant (uniform in space), then

$$-\Delta \rho = \frac{1}{k} f$$

where

$$\Delta \rho = \nabla \cdot \nabla \rho = \frac{\partial^2 \rho}{\partial x^2} + \frac{\partial^2 \rho}{\partial y^2} + \frac{\partial^2 \rho}{\partial z^2}$$

$\Delta = \nabla \cdot \nabla$ is called the **Laplace operator**.

Electrostatic potential

By definition $\mathbf{E} = -\nabla\phi$, where ϕ is the electrostatic potential. The Gauss law and the Divergence Theorem give

$$Q = \iiint_{\Omega} \rho dV = \varepsilon_0 \iint_S \mathbf{E} \cdot d\mathbf{S} = \varepsilon_0 \iiint_{\Omega} \nabla \cdot \mathbf{E} dV = -\varepsilon_0 \iiint_{\Omega} \nabla \cdot \nabla \phi dV$$

where ρ is the charge density. Hence, ϕ satisfies the Poisson equation

$$-\Delta\phi = \frac{1}{\varepsilon_0}\rho$$

Gradient of a vector

In some equations we meet:

$$\nabla \mathbf{u} = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} \begin{bmatrix} u_x & u_y & u_z \end{bmatrix} = \begin{bmatrix} \frac{\partial u_x}{\partial x} & \frac{\partial u_y}{\partial x} & \frac{\partial u_z}{\partial x} \\ \frac{\partial u_x}{\partial y} & \frac{\partial u_y}{\partial y} & \frac{\partial u_z}{\partial y} \\ \frac{\partial u_x}{\partial z} & \frac{\partial u_y}{\partial z} & \frac{\partial u_z}{\partial z} \end{bmatrix}$$

Or the nonlinear term:

$$\begin{aligned} \mathbf{u} \cdot \nabla \mathbf{u} &= \begin{bmatrix} u_x & u_y & u_z \end{bmatrix} \begin{bmatrix} \frac{\partial u_x}{\partial x} & \frac{\partial u_y}{\partial x} & \frac{\partial u_z}{\partial x} \\ \frac{\partial u_x}{\partial y} & \frac{\partial u_y}{\partial y} & \frac{\partial u_z}{\partial y} \\ \frac{\partial u_x}{\partial z} & \frac{\partial u_y}{\partial z} & \frac{\partial u_z}{\partial z} \end{bmatrix} \\ &= \begin{bmatrix} u_x \frac{\partial u_x}{\partial x} + u_y \frac{\partial u_x}{\partial y} + u_z \frac{\partial u_x}{\partial z}, & u_x \frac{\partial u_y}{\partial x} + u_y \frac{\partial u_y}{\partial y} + u_z \frac{\partial u_y}{\partial z}, \\ u_x \frac{\partial u_z}{\partial x} + u_y \frac{\partial u_z}{\partial y} + u_z \frac{\partial u_z}{\partial z} \end{bmatrix} \end{aligned}$$

Divergence of a tensor

In elasticity the stress $\mathbb{T}$ is a tensor:

$$\mathbb{T} = \begin{bmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{bmatrix}$$

The divergence of a tensor is defined as

$$\begin{aligned} \nabla \cdot \mathbb{T} &= \begin{bmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \end{bmatrix} \begin{bmatrix} T_{xx} & T_{xy} & T_{xz} \\ T_{yx} & T_{yy} & T_{yz} \\ T_{zx} & T_{zy} & T_{zz} \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial T_{xx}}{\partial x} + \frac{\partial T_{yx}}{\partial y} + \frac{\partial T_{zx}}{\partial z}, & \frac{\partial T_{xy}}{\partial x} + \frac{\partial T_{yy}}{\partial y} + \frac{\partial T_{zy}}{\partial z}, \\ \frac{\partial T_{xz}}{\partial x} + \frac{\partial T_{yz}}{\partial y} + \frac{\partial T_{zz}}{\partial z} \end{bmatrix} \end{aligned}$$